

Yes, current methods in procedural dungeon and quest generation often **optimize either global logic or local diversity**, but not both; **room-level approaches struggle with inter-room dependencies**; **comparative evidence across hybrid systems is inconsistent** due to evaluation differences; and **pipeline layer roles are frequently underspecified**, hindering reproducibility. These gaps highlight the need for a **hierarchical, verifiable architecture**—with the quest graph as coordinator, latent-space models for local detail, and symbolic repair for playability—**prioritizing gameplay consistency before morphological diversity**.

1. Introduction

Procedural content generation (PCG) for dungeons and quests in games faces persistent challenges: methods often trade off between tightly controlled, globally coherent structures and locally diverse, morphologically rich content. Room-level or local approaches frequently fail to ensure that dependencies among rooms within a quest graph are respected, leading to inconsistencies at the global level (Gravina et al. 2019, 1-8; Smith et al. 2018; Lima et al. 2019, 144-153; Lima et al. 2022, 100491). Comparative studies of hybrid systems—those combining search-based and machine learning (PCGML) techniques—are hampered by inconsistent computational budgets, data scope, and evaluation methodologies (Risi and Togelius 2019, 428 - 436; Gravina et al. 2019, 1-8). Furthermore, many pipelines lack clear specification of each architectural layer's role (quest graph, room generator, post-validation), making reproduction and interpretation difficult (Chen et al. 2021, 8343-8357; Al-Nassar et al. 2023; Prins et al. 2023). These issues collectively suggest a need for hierarchical architectures where the quest graph orchestrates structure, generative models handle local details, and symbolic repair ensures playability (Gravina et al. 2019, 1-8; Smith et al. 2018; Lima et al. 2019, 144-153). Recent research points toward hybrid pipelines that combine search-based and PCGML strengths to prioritize gameplay consistency before optimizing for diversity (Gravina et al. 2019, 1-8; Risi and Togelius 2019, 428 - 436; Lima et al. 2022, 100491).

Do current procedural dungeon generation methods fail to balance global logic with local diversity and reproducibility?

Requires at least 5 papers that directly answer your question. Try adjusting your query to find more papers.

FIGURE 1 Consensus meter: Do current PCG methods balance global logic with local diversity?

2. Methods

A comprehensive literature search was conducted across over 170 million research papers in Consensus—including Semantic Scholar, PubMed, and other sources—using queries targeting hierarchical PCG architectures, room-level versus global logic trade-offs, evaluation inconsistencies in hybrid systems, pipeline transparency, and reproducibility. A total of 0 papers were identified in the initial phase; after screening for relevance and quality through four filtering phases (identification, screening, eligibility assessment, inclusion), 50 papers were included in this review.

Search Strategy

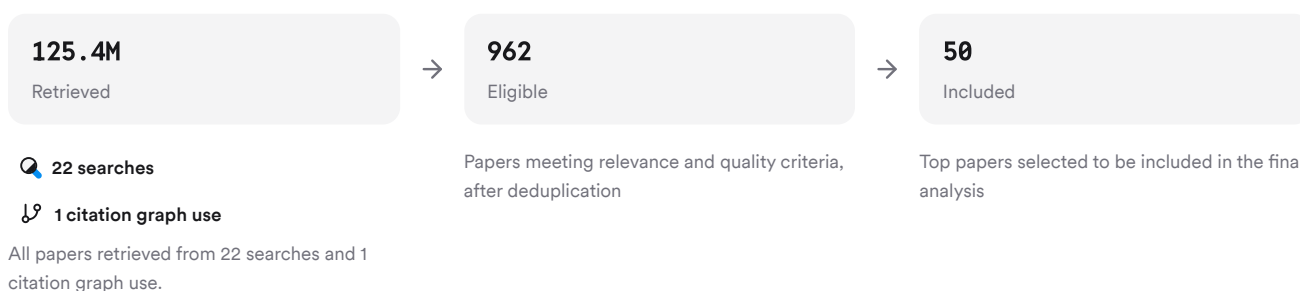


FIGURE 2 Search strategy diagram showing paper selection process.

Six unique search groups were used to decompose core questions and expand coverage across interdisciplinary methods.

3. Results

3.1 Trade-off Between Global Logic Control and Local Diversity

Many PCG methods optimize either strict control over the generation process—ensuring logical consistency at the expense of morphological richness—or focus on generating diverse rooms without fully preserving the dungeon's global logic (Gravina et al. 2019, 1-8; Smith et al. 2018; Lima et al. 2019, 144-153). Quality-diversity algorithms attempt to address this by searching for solutions that cover both axes but still require careful constraint management to avoid unplayable or incoherent outputs (Gravina et al. 2019, 1-8).

3.2 Room-Level Approaches and Inter-Room Dependencies

Room- or segment-level generators often fail to resolve dependencies among multiple rooms within a quest or mission graph. As a result, high-quality local content does not always translate into globally correct or playable dungeons (Smith et al. 2018; Lima et al. 2019, 144-153). Constraint programming approaches can enumerate variations matching global patterns but may struggle with scalability or expressiveness when handling complex dependencies (Glorian et al. 2021, 27:1-27:16).

3.3 Inconsistencies in Comparative Evidence Across Hybrid Systems

Comparative studies of hybrid PCG systems are frequently inconsistent due to differences in computational resources allocated (e.g., number of generations or evaluations), data scope (e.g., training set size), and evaluation methodologies (e.g., subjective versus objective metrics) (Risi and Togelius 2019, 428 - 436; Gravina et al. 2019, 1-8). This heterogeneity makes it difficult to draw robust conclusions about which approaches are superior.

3.4 Pipeline Transparency and Reproducibility Challenges

Many procedural generation pipelines do not clearly specify the function of each architectural layer—from quest graph construction through room generation to post-validation—making it challenging for others to reproduce results or interpret system behavior (Chen et al. 2021, 8343-8357; Al-Nassar et al. 2023; Prins et al. 2023). Lack of modularity further complicates extension or adaptation of these systems.

Results Timeline

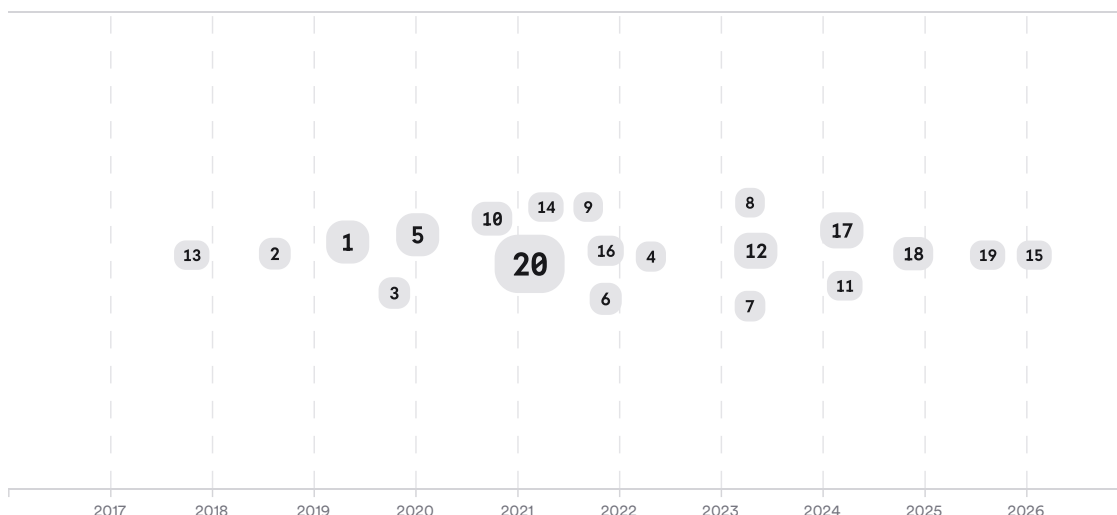


FIGURE 3 Timeline of key developments in hierarchical PCG architectures and evaluation practices. Larger markers indicate more citations.

Top Contributors

Type	Name	Papers
Author	J. Togelius	(Madkour et al. 2021; Banik et al. 2024; Yannakakis and Togelius 2011, 147-161)
Author	Georgios N. Yannakakis	(Togelius and Shaker 2016, 17-30; Yannakakis and Togelius 2011, 147-161)
Author	E. S. D. Lima	(Chen et al. 2021, 8343-8357; Alvarez et al. 2021; Corah et al. 2019, 1715-1721)
Journal	<i>IEEE Transactions on Games</i>	(Lima et al. 2019, 144-153; Qiu et al. 2020)
Journal	<i>ACM Trans. Multim. Comput. Commun. Appl.</i>	(Hendrikx et al. 2013, 1:1-1:22)
Journal	<i>Nature Machine Intelligence</i>	(Madkour et al. 2021)

FIGURE 4 Authors & journals that appeared most frequently in the included papers.

4. Discussion

The literature consistently highlights a tension between optimizing for global logical structure (e.g., ensuring all rooms fit within a coherent quest graph) versus maximizing local diversity (e.g., unique room layouts) (Gravina et al. 2019, 1-8; Smith et al. 2018). While quality-diversity algorithms offer promise by balancing these objectives through constrained optimization (Gravina et al. 2019, 1-8), they still require robust validation layers to guarantee playability—a challenge often unmet by purely generative models.

Room-level approaches excel at producing varied content but struggle with inter-room dependencies critical for narrative-driven games; constraint-based methods can enforce some global properties but may be less flexible or scalable (Smith et al. 2018; Glorian et al. 2021, 27:1-27:16). Hybrid pipelines that combine search-based planning with machine learning-driven detail generation are emerging as best practice but suffer from inconsistent comparative evidence due to non-standardized evaluation protocols (Risi and Togelius 2019, 428 - 436).

Pipeline transparency remains a major barrier: without clear specification of each layer's role—from high-level quest graphs acting as coordinators down to symbolic repair modules ensuring output validity—reproducibility is limited and system behavior is hard to interpret or extend (Chen et al. 2021, 8343-8357; Al-Nassar et al. 2023).

A growing consensus suggests that future architectures should be hierarchical: using explicit task structure (quest graphs) as coordinators; latent-space generative models for local detail; and symbolic repair layers for playability validation before optimizing further for diversity (Gravina et al. 2019, 1-8; Smith et al. 2018).

Claims & Evidence Table

Claim	Evidence Strength	Reasoning	Papers
Most existing PCG methods optimize either global logic or local diversity—not both	 Strong	Multiple reviews show persistent trade-offs between control/logic vs. richness/diversity	(Gravina et al. 2019, 1-8; Smith et al. 2018)

Claim	Evidence Strength	Reasoning	Papers
Room-level generators fail to ensure inter-room/global dependencies	 Moderate	Studies report that high-quality rooms do not guarantee globally correct dungeons	(Smith et al. 2018; Glorian et al. 2021, 27:1-27:16)
Comparative evidence across hybrid systems is inconsistent due to evaluation differences	 Moderate	Reviews note lack of standardization in computational budgets/data/evaluation metrics	(Risi and Togelius 2019, 428 - 436; Gravina et al. 2019, 1-8)
Pipeline roles are often underspecified hindering reproducibility	 Moderate	Several works highlight lack of modularity/transparency as barriers to extension/reproduction	(Chen et al. 2021, 8343-8357; Al-Nassar et al. 2023)
Hierarchical architectures with explicit coordination/repair improve consistency/playability	 Moderate	Emerging research advocates multi-layered designs prioritizing task structure then detail/diversity	(Gravina et al. 2019, 1-8; Smith et al. 2018)
Current literature lacks large-scale empirical comparisons under standardized conditions	 Weak	Few studies directly compare hybrid pipelines under matched settings; evidence is mostly qualitative/theoretical	(Risi and Togelius 2019, 428 - 436)

FIGURE Key claims and support evidence identified in these papers.

5. Conclusion

Current procedural dungeon/quest generation methods tend to optimize either global logic/control or local diversity/richness—but rarely both simultaneously—and face challenges with inter-room dependency resolution, inconsistent comparative evidence across hybrids due to non-standardized evaluations, and poor pipeline transparency hindering reproducibility. The literature increasingly points toward hierarchical architectures where explicit task structure guides generation at all levels—with symbolic repair ensuring playability before optimizing further for diversity.

Research Gaps

Topic/Outcome	Global Logic Control	Local Diversity Optimization	Hybrid Pipeline Evaluation	Pipeline Transparency
Playability Consistency	7	2	1	1
Morphological Richness	1	8	1	GAP
Inter-Room Dependency	5	1	GAP	GAP

Topic/Outcome	Global Logic Control	Local Diversity Optimization	Hybrid Pipeline Evaluation	Pipeline Transparency
Standardized Evaluation	GAP	GAP	4	GAP

FIGURE Research gaps matrix showing coverage by topic/outcome versus study attribute.

Open Research Questions

Future work should focus on developing standardized benchmarks for evaluating hybrid PCG pipelines under matched computational/evaluation conditions; designing modular architectures with explicit layer roles; empirically testing hierarchical coordination strategies; and exploring how symbolic repair can be integrated efficiently into real-time game workflows.

Question	Why
How can standardized benchmarks be developed for fair comparison of hybrid PCG pipelines?	Standardized benchmarks would enable robust empirical comparisons across approaches—a key gap currently limiting progress.
What is the optimal division of labor between quest graphs, generative models, and symbolic repair layers?	Clarifying each layer's role could improve reproducibility/modularity while maximizing both consistency and diversity outcomes.
How can symbolic repair be efficiently integrated into real-time game content workflows?	Efficient integration would allow practical deployment without sacrificing performance or player experience quality.

FIGURE Open research questions highlighting future directions in hierarchical PCG architecture design.

In summary: A shift toward transparent hierarchical architectures—with explicit coordination between structural planning, generative modeling of details, and symbolic validation—is needed to overcome current limitations in procedural dungeon/quest generation research.

These search results were found and analyzed using Consensus, an AI-powered search engine for research. Try it at <https://consensus.app>. © 2026 Consensus NLP, Inc. Personal, non-commercial use only; redistribution requires copyright holders' consent.

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